



DATA MINING COURSE (IS471)



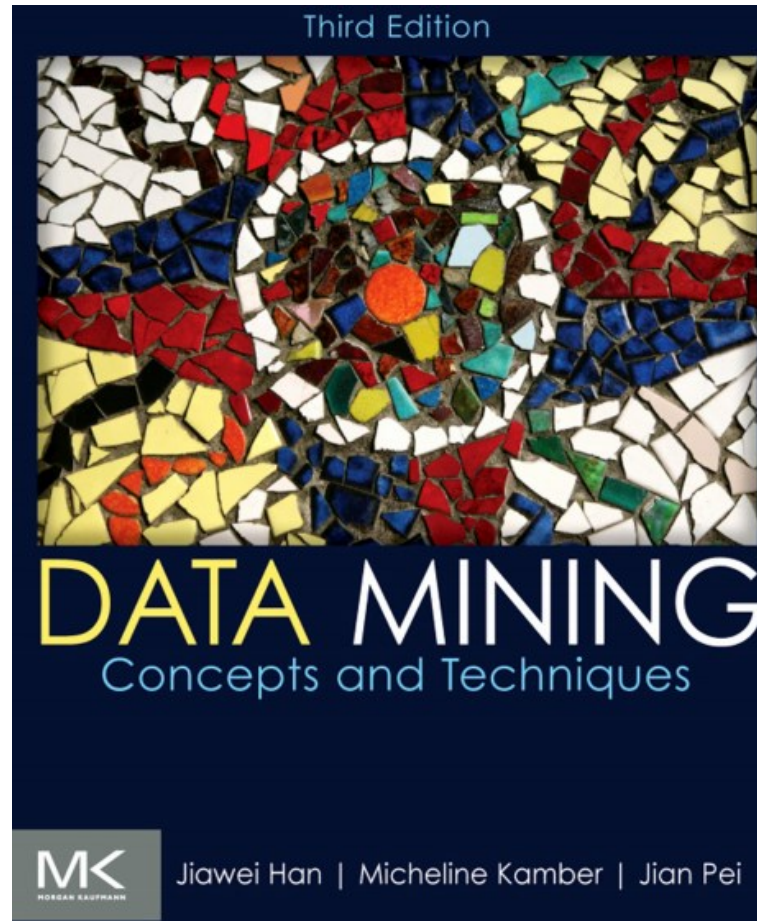
About the course – Administtrivia Instructor:

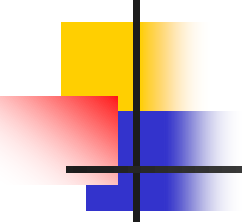
Dr. Osama Farouk Hassan

dr.osama.farouk.is@ci.suez.edu.eg

IS471, Sunday 2:00-5:00 pm (online)

Text Book:



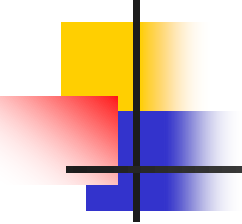


Assessment Methods

- **Class Work** : Quiz + Assignment during the term

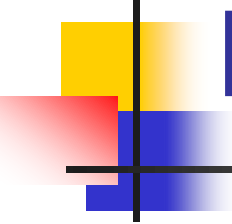
Degree	Item
5	Quiz1
5	Quiz2
2	Assignment1
2	Assignment2
2	Assignment3
2	Assignment4
2	Assignment5
Total	20
Oral and Lab.	20

- **Final exam** (80).
- **Total** (120).



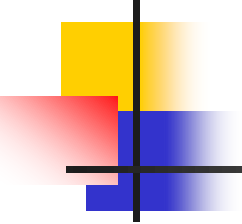
Data Mining

- **Lecture 1 - Introduction**



Lecture Outlines

- What is Data Mining?
- Knowledge Discovery from Data (KDD)
- Examples of Large Datasets
- Examples of Data mining Applications
- How Data Mining is used
- The Data Mining Process
- What Kinds of Data Can Be Mined?
- Why Data Preprocessing?
- What Kinds of Patterns Can Be Mined?
- Which Technologies are Used?



What is Data Mining?

- **Data Mining is:**

The process of discovering interesting patterns and knowledge from large amounts of data. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.

What Is Data Mining?

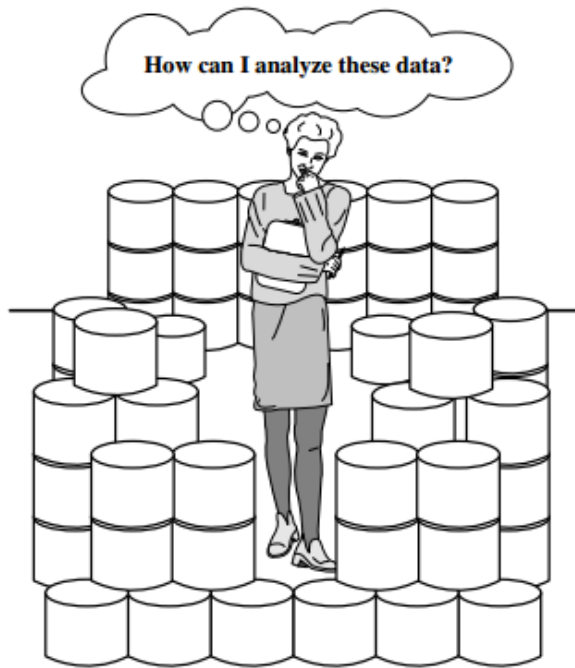


Figure 1.2 The world is data rich but information poor.

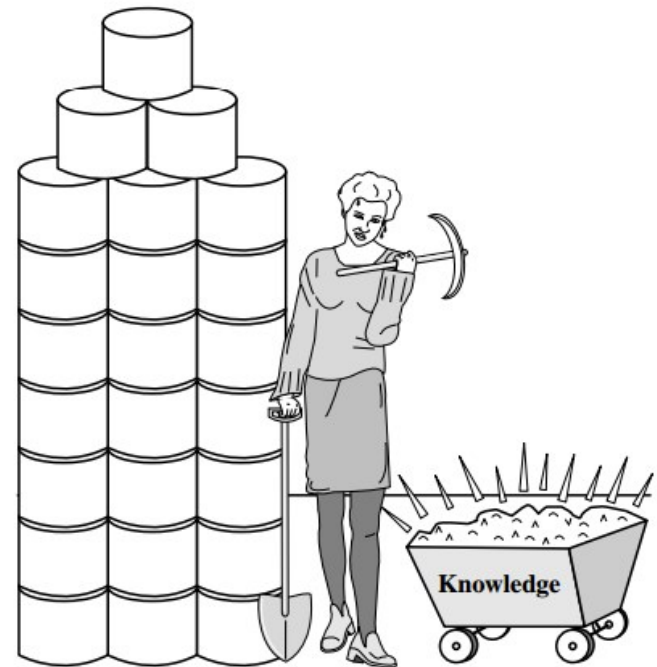
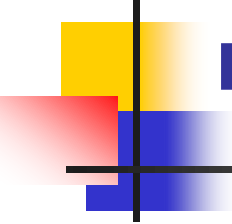


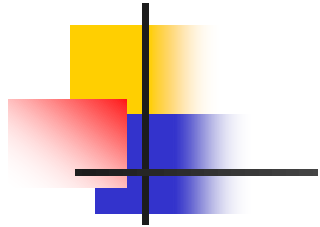
Figure 1.3 Data mining—searching for knowledge (interesting patterns) in data.



Knowledge Discovery from Data (KDD)

The knowledge discovery process is shown in Figure 1.4 as an iterative sequence of the following steps:

- 1) **Data cleaning** (to remove noise and inconsistent data)
- 2) **Data integration** (where multiple data sources may be combined)
- 3) **Data selection** (where data relevant to the analysis task are retrieved from the database)
- 4) **Data transformation** (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
- 5) **Data mining** (an essential process where intelligent methods are applied to extract data patterns)
- 6) **Pattern evaluation** (to identify the truly interesting patterns representing knowledge based on interestingness measures.)
- 7) **Knowledge presentation** (where visualization and knowledge representation techniques are used to present mined knowledge to users)



Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.

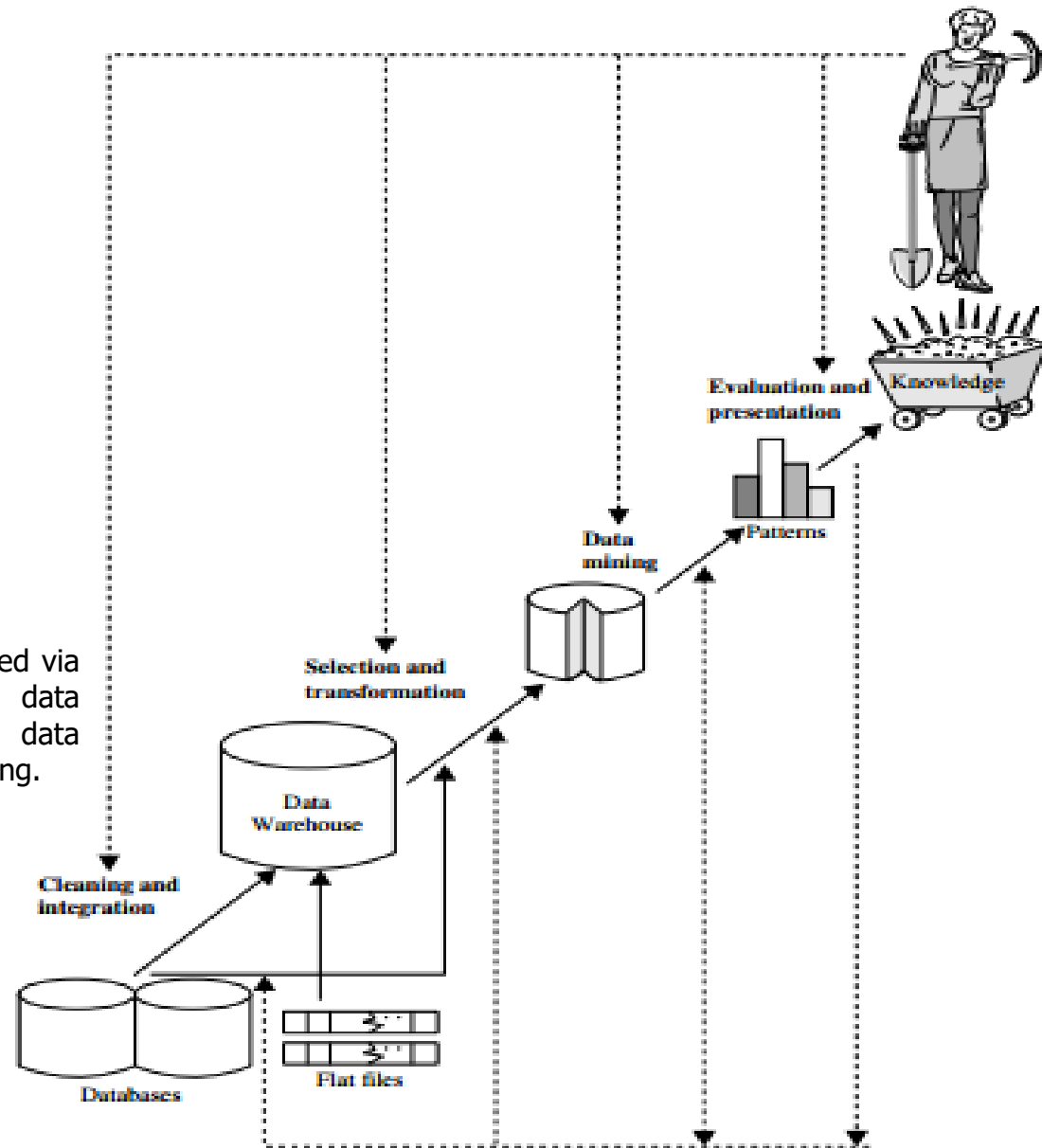
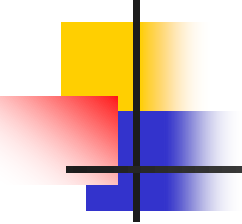
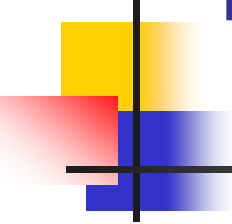


Figure 1.4 Data mining as a step in the process of knowledge discovery.



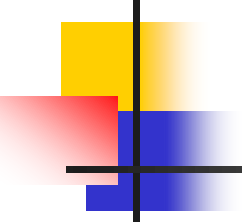
Examples of Large Datasets

- Government: IRS, NGA, ...
- Large corporations
 - WALMART: 20M transactions per day
 - MOBIL: 100 TB geological databases
 - AT&T 300 M calls per day
 - Credit card companies
- Scientific
 - NASA, EOS project: 50 GB per hour
 - Environmental datasets



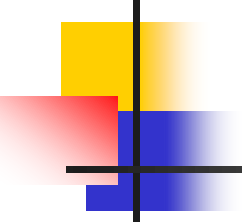
Examples of Data mining Applications

1. Fraud detection: credit cards, phone cards
2. Marketing: customer targeting
3. Data Warehousing: WalMart
4. Astronomy
5. biology



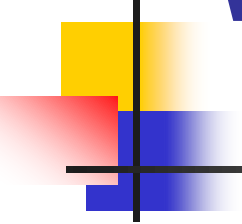
How Data Mining is used

1. Identify the problem
2. Use data mining techniques to transform the data into information
3. Act on the information
4. Measure the results



The Data Mining Process

1. Understand the domain
2. Create a dataset:
 - Select the interesting attributes
 - Data cleaning and preprocessing
3. Choose the data mining task and the specific algorithm
4. Interpret the results, and possibly return to 2



What Kinds of Data Can Be Mined?

- Database Data.
- Data Warehouses.
- Transactional Data.
- Other Kinds of Data.

Database Data

- A **database system**, also called a **database management system** (DBMS), consists of a collection of interrelated data, known as a database, and a set of software programs to manage and access the data.
- A relational database is a collection of tables, each of which is assigned a unique name. Each table consists of a set of attributes (columns or fields) and usually stores a large set of tuples (records or rows). Each tuple in a relational table represents an object identified by a unique key and described by a set of attribute values. A semantic data model, such as an **entity-relationship (ER)** data model, is often constructed for relational databases. An ER data model represents the database as a set of entities and their relationships.

Example 1.2 A relational database for AllElectronics

<i>customer</i>	(<i>cust_ID</i> , <i>name</i> , <i>address</i> , <i>age</i> , <i>occupation</i> , <i>annual_income</i> , <i>credit_information</i> , <i>category</i> , ...)
<i>item</i>	(<i>item_ID</i> , <i>brand</i> , <i>category</i> , <i>type</i> , <i>price</i> , <i>place_made</i> , <i>supplier</i> , <i>cost</i> , ...)
<i>employee</i>	(<i>empl_ID</i> , <i>name</i> , <i>category</i> , <i>group</i> , <i>salary</i> , <i>commission</i> , ...)
<i>branch</i>	(<i>branch_ID</i> , <i>name</i> , <i>address</i> , ...)
<i>purchases</i>	(<i>trans_ID</i> , <i>cust_ID</i> , <i>empl_ID</i> , <i>date</i> , <i>time</i> , <i>method_paid</i> , <i>amount</i>)
<i>items_sold</i>	(<i>trans_ID</i> , <i>item_ID</i> , <i>qty</i>)
<i>works_at</i>	(<i>empl_ID</i> , <i>branch_ID</i>)

Figure 1.5 Relational schema for a relational database, *AllElectronics*.

Data Warehouses

Suppose that AllElectronics is a successful international company with branches around the world. Each branch has its own set of databases.

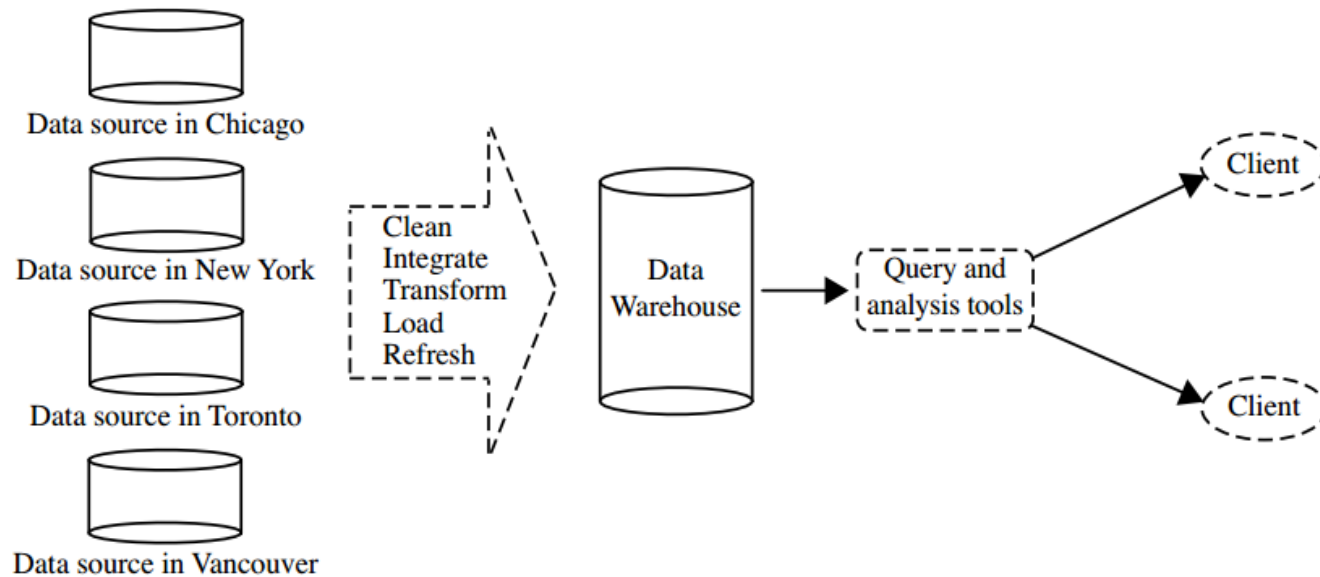


Figure 1.6 Typical framework of a data warehouse for *AllElectronics*.

Example 1.3 A data cube for AllElectronics.

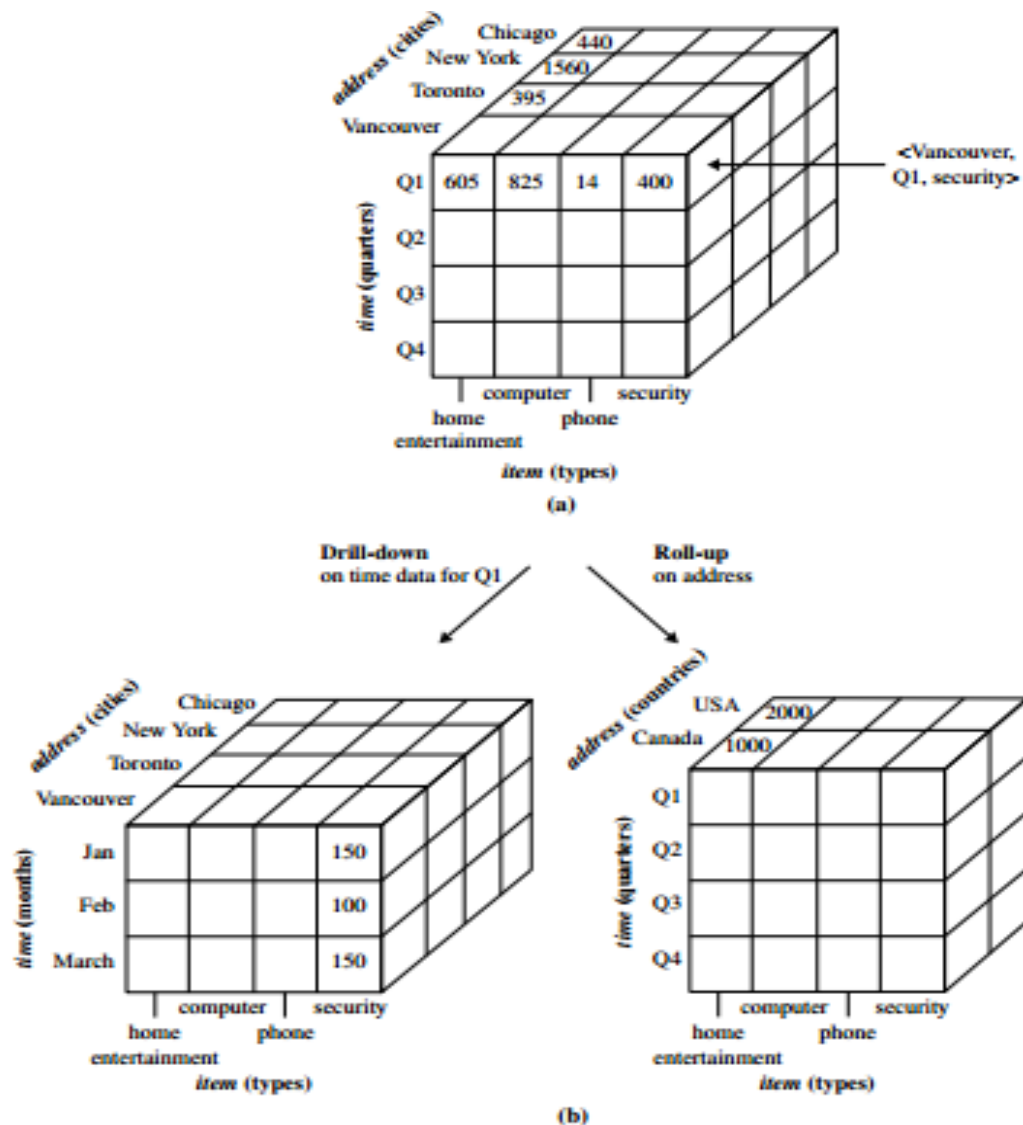
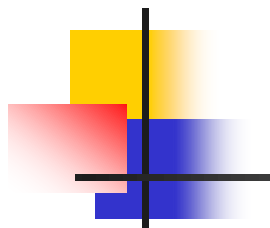


Figure 1.7 A multidimensional data cube, commonly used for data warehousing, (a) showing summarized data for AllElectronics and (b) showing summarized data resulting from drill-down and roll-up operations on the cube in (a). For improved readability, only some of the cube cell values are shown.

Transactional Data

The transactional database is usually either stored in a flat file in a format similar to the table in Figure 1.8 or unfolded into a standard relation in a format similar to the items sold table in Figure 1.5.

<i>trans_ID</i>	<i>list_of_item_IDs</i>
T100	I1, I3, I8, I16
T200	I2, I8
...	...

Figure 1.8 Fragment of a transactional database for sales at *AllElectronics*.

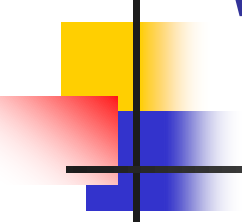
<i>customer</i>	(<i>cust_ID</i> , <i>name</i> , <i>address</i> , <i>age</i> , <i>occupation</i> , <i>annual_income</i> , <i>credit_information</i> , <i>category</i> , ...)
<i>item</i>	(<i>item_ID</i> , <i>brand</i> , <i>category</i> , <i>type</i> , <i>price</i> , <i>place_made</i> , <i>supplier</i> , <i>cost</i> , ...)
<i>employee</i>	(<i>empl_ID</i> , <i>name</i> , <i>category</i> , <i>group</i> , <i>salary</i> , <i>commission</i> , ...)
<i>branch</i>	(<i>branch_ID</i> , <i>name</i> , <i>address</i> , ...)
<i>purchases</i>	(<i>trans_ID</i> , <i>cust_ID</i> , <i>empl_ID</i> , <i>date</i> , <i>time</i> , <i>method_paid</i> , <i>amount</i>)
<i>items_sold</i>	(<i>trans_ID</i> , <i>item_ID</i> , <i>qty</i>)
<i>works_at</i>	(<i>empl_ID</i> , <i>branch_ID</i>)

Figure 1.5 Relational schema for a relational database, *AllElectronics*.



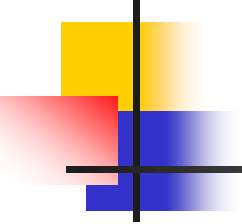
Other Kinds of Data

Such kinds of data can be seen in many applications: time-related or sequence data (e.g., historical records, stock exchange data, and time-series and biological sequence data), data streams (e.g., video surveillance and sensor data, which are continuously transmitted), spatial data (e.g., maps), engineering design data (e.g., the design of buildings, system components, or integrated circuits), hypertext and multimedia data (**including text, image, video, and audio data**), graph and networked data (e.g., social and information networks), and the Web (a huge, widely distributed information repository made available by the Internet).



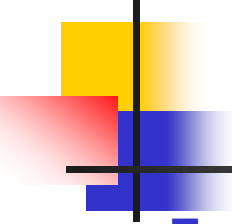
Why Data Preprocessing?

- **Data in the real world is dirty**
 - **incomplete**: lacking *attribute values*, lacking certain *attributes of interest*, or containing only aggregate data
 - **noisy**: containing errors or outliers
 - **inconsistent**: containing discrepancies in codes or names
- **No quality data, no quality mining results!**
 - Quality decisions must be based on quality data
 - Data warehouse needs consistent integration of quality data
 - Required for both OLAP and Data Mining!



Why can Data be Incomplete?

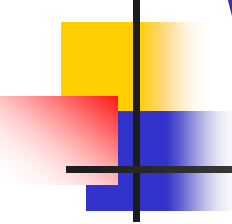
- Attributes of interest are not available (e.g., customer information for sales transaction data)
- Data were not considered important at the time of transactions, so they were not recorded!
- Data not recorder because of misunderstanding or malfunctions
- Data may have been recorded and later deleted!
- Missing/unknown values for some data



Data Cleaning

- Data cleaning tasks

- Fill in missing values
- Identify outliers and smooth out noisy data
- Correct inconsistent data



What Kinds of Patterns Can Be Mined?

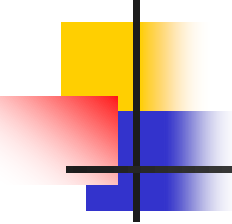
In general, such tasks can be classified into two categories:

Descriptive mining: tasks characterize properties of the data in a target data set.

Predictive mining: tasks perform induction on the current data in order to make predictions.

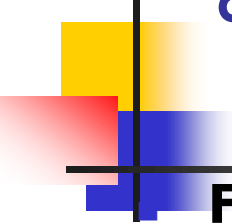
- Class/Concept Description: Characterization and Discrimination.
- Mining Frequent Patterns, Associations, and Correlations.
- Classification and Regression for Predictive Analysis.
- Cluster Analysis.
- Outlier Analysis .

Class/Concept Description: Characterization and Discrimination



- Data entries can be associated with classes or concepts. For example, in the AllElectronics store, classes of items for sale include computers and printers, and concepts of customers include big Spenders and budget Spenders.
- **Data characterization** is a summarization of the general characteristics or features of a target class of data.
- **Data discrimination** is a comparison of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.

Mining Frequent Patterns, Associations, and Correlations



Frequent patterns, as the name suggests, are patterns that occur frequently in data. There are many kinds of frequent patterns.

A frequent itemset typically refers to a set of items that often appear together in a transactional data set—for example, milk and bread, which are frequently bought together in grocery stores by many customers.

A frequently occurring subsequence, such as the pattern that customers tend to purchase first a laptop, followed by a digital camera, and then a memory card, is a (frequent) sequential pattern.

A frequently substructure can refer to different structural forms (e.g., graphs, trees, or lattices) that may be combined with itemsets or subsequences.

Mining frequent patterns leads to the discovery of interesting associations and correlations within data.

Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection;
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

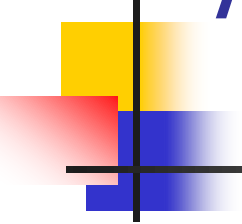
<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

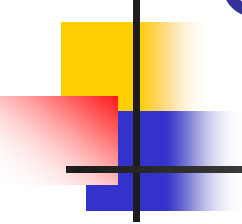
{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}

Association Rule Discovery: Application 1



- Marketing and Sales Promotion:
 - Let the rule discovered be
 $\{Bagels, \dots\} \rightarrow \{Potato\ Chips\}$
 - Potato Chips as consequent => Can be used to determine what should be done to boost its sales.
 - Bagels in the antecedent => Can be used to see which products would be affected if the store discontinues selling bagels.
 - Bagels in antecedent *and* Potato chips in consequent => Can be used to see what products should be sold with Bagels to promote sale of Potato chips!



Classification: Definition

- Given a collection of records (*training set*)
 - Each record contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- Goal: previously unseen records should be assigned a class as accurately as possible.
 - A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

Classification Example

categorical

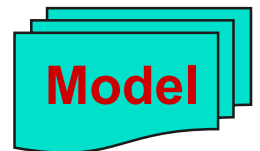
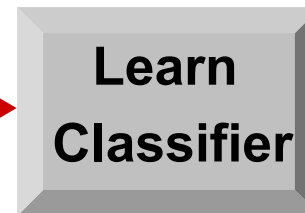
categorical

continuous

class

Tid	Home Owner	Marital Status	Taxable Income	Default
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Home Owner	Marital Status	Taxable Income	Default
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?

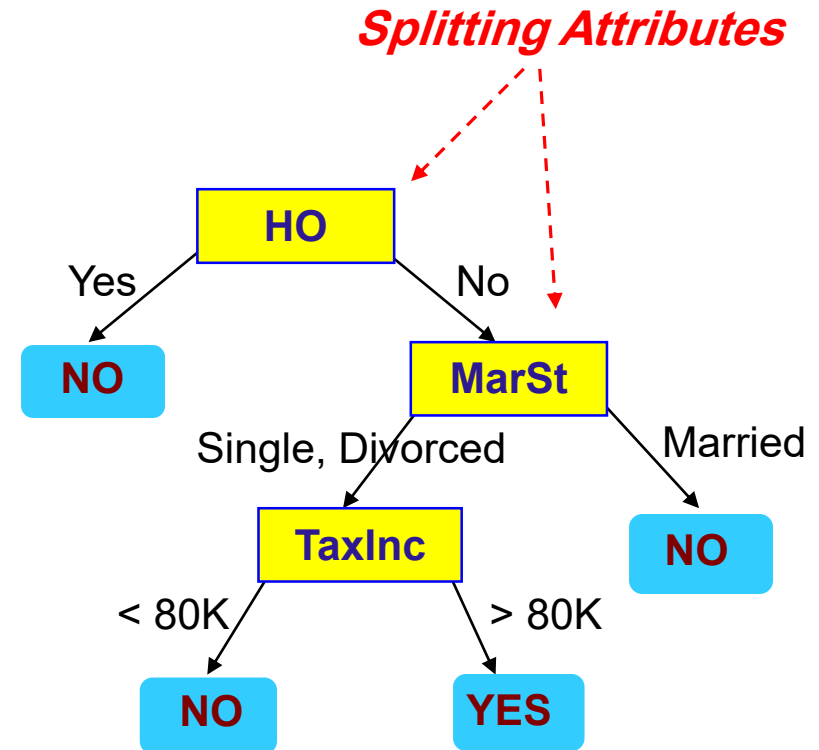


Example of a Decision Tree

categorical
categorical
continuous
class

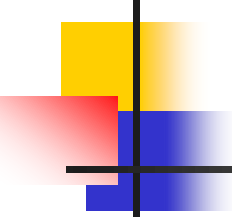
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10	No	Single	90K	Yes

Training Data



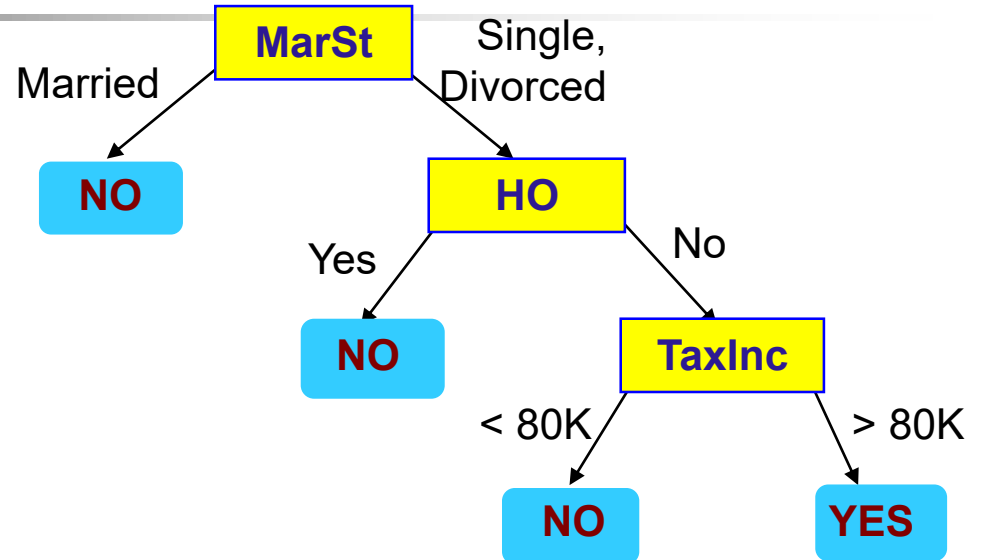
Model: Decision Tree

Another Example of Decision Tree

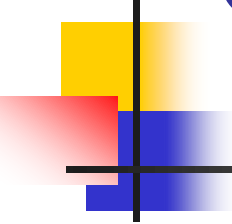


categorical *categorical* *continuous* *class*

<i>Tid</i>	Home Owner	Marital Status	Taxable Income	Default
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6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

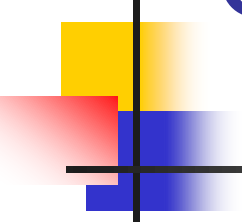


There could be more than one tree that fits the same data!



Classification: Application 1

- Direct Marketing
 - Goal: Reduce cost of mailing by *targeting* a set of consumers likely to buy a new cell-phone product.
 - Approach:
 - Use the data for a similar product introduced before.
 - We know which customers decided to buy and which decided otherwise. This *{buy, don't buy}* decision forms the *class attribute*.
 - Collect various demographic, lifestyle, and company-interaction related information about all such customers.
 - Type of business, where they stay, how much they earn, etc.
 - Use this information as input attributes to learn a classifier model.

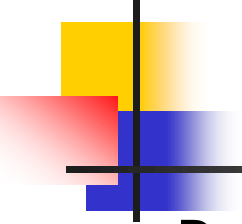


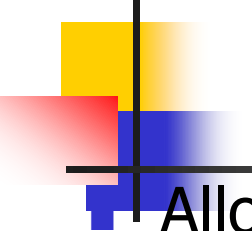
Classification: Application 2

- Fraud Detection

- Goal: Predict fraudulent cases in credit card transactions.
- Approach:
 - Use credit card transactions and the information on its account-holder as attributes.
 - When does a customer buy, what does he buy, how often he pays on time, etc
 - Label past transactions as fraud or fair transactions. This forms the class attribute.
 - Learn a model for the class of the transactions.
 - Use this model to detect fraud by observing credit card transactions on an account.

Clustering

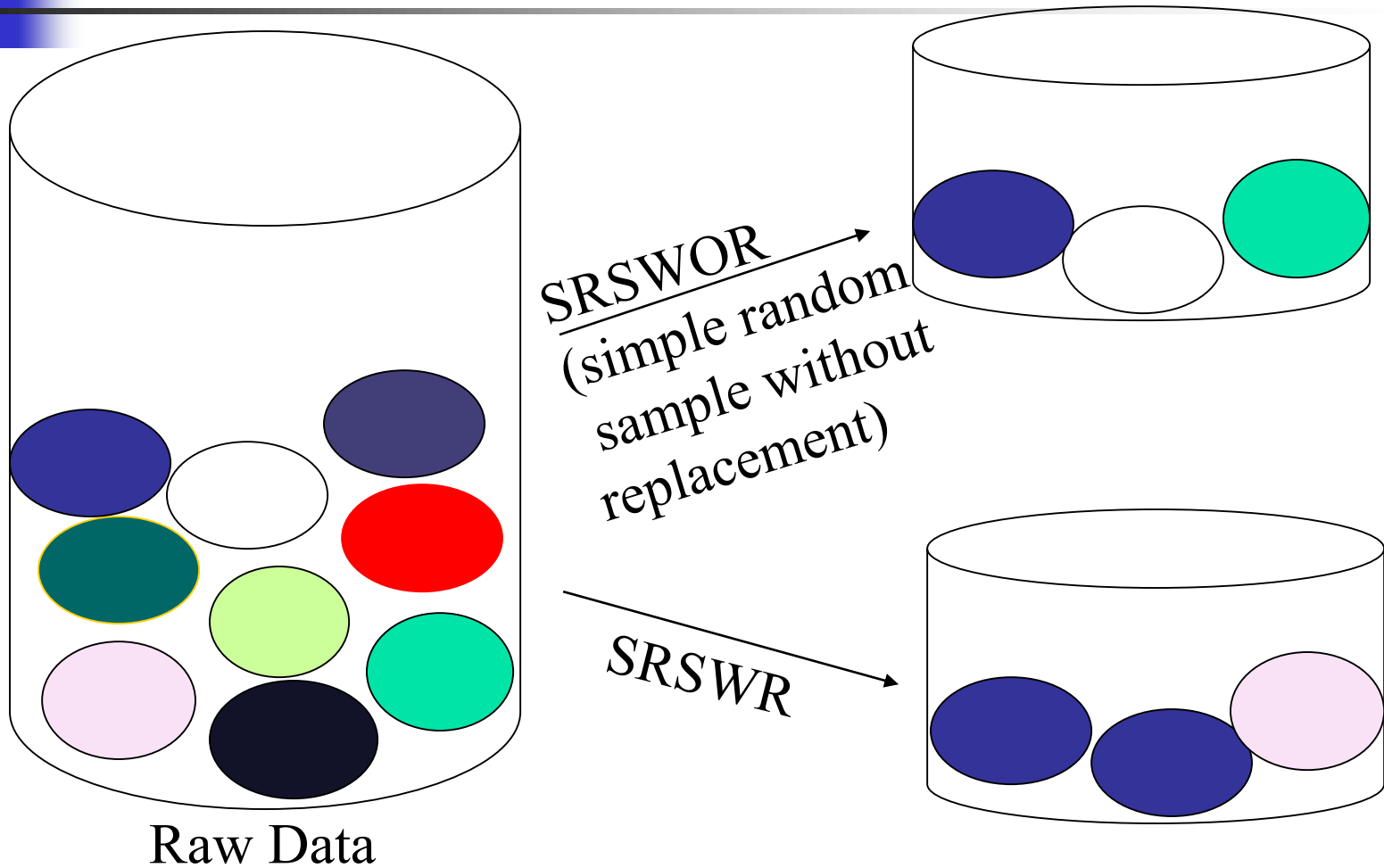
- 
-
- Partitions data set into clusters, and models it by one representative from each cluster
 - Can be very effective if data is clustered but not if data is “smeared”
 - There are many choices of clustering definitions and clustering algorithms, more later!



Sampling

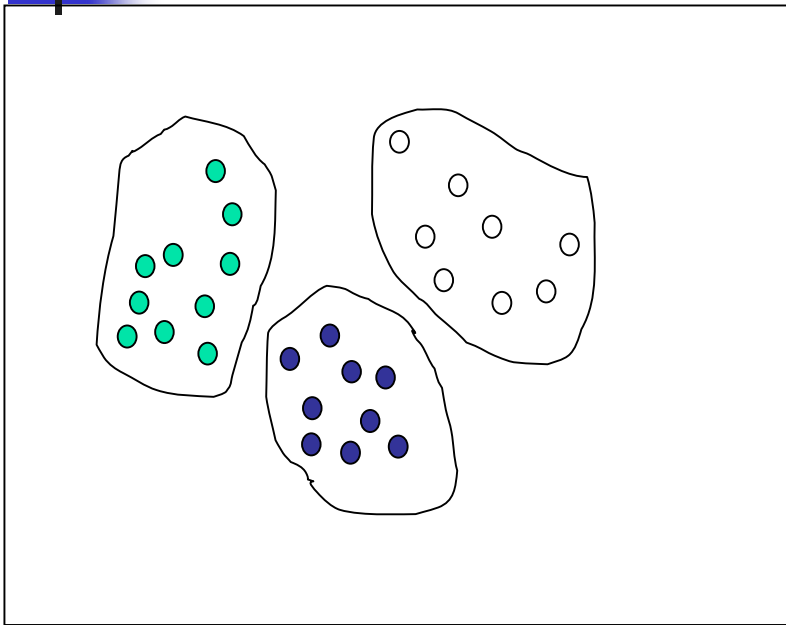
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
- Develop adaptive sampling methods
 - Stratified sampling:
 - Approximate the percentage of each class (or subpopulation of interest) in the overall database
 - Used in conjunction with skewed data
- Sampling may not reduce database I/Os (page at a time).

Sampling

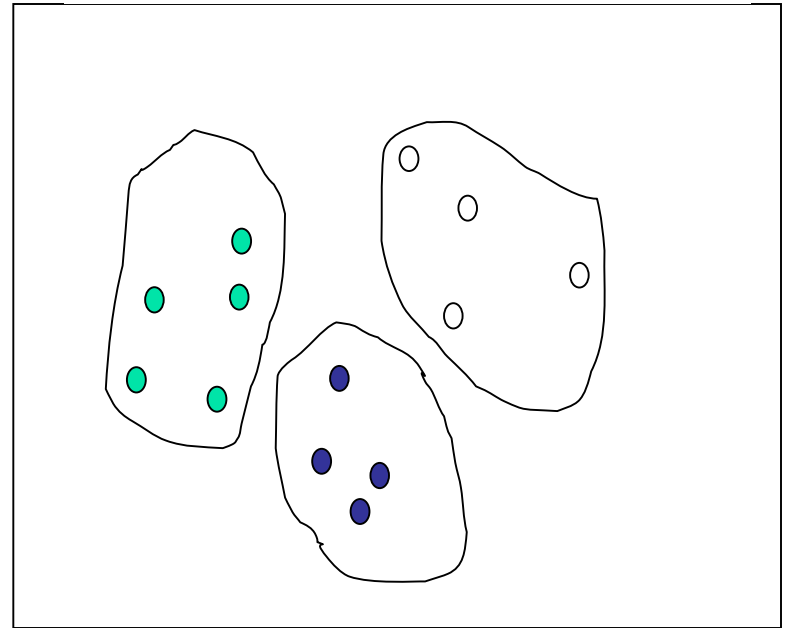


Sampling

Raw Data



Cluster/Stratified Sample



- The number of samples drawn from each cluster/stratum is analogous to its size
- Thus, the samples represent better the data and outliers are avoided

Which Technologies Are Used?

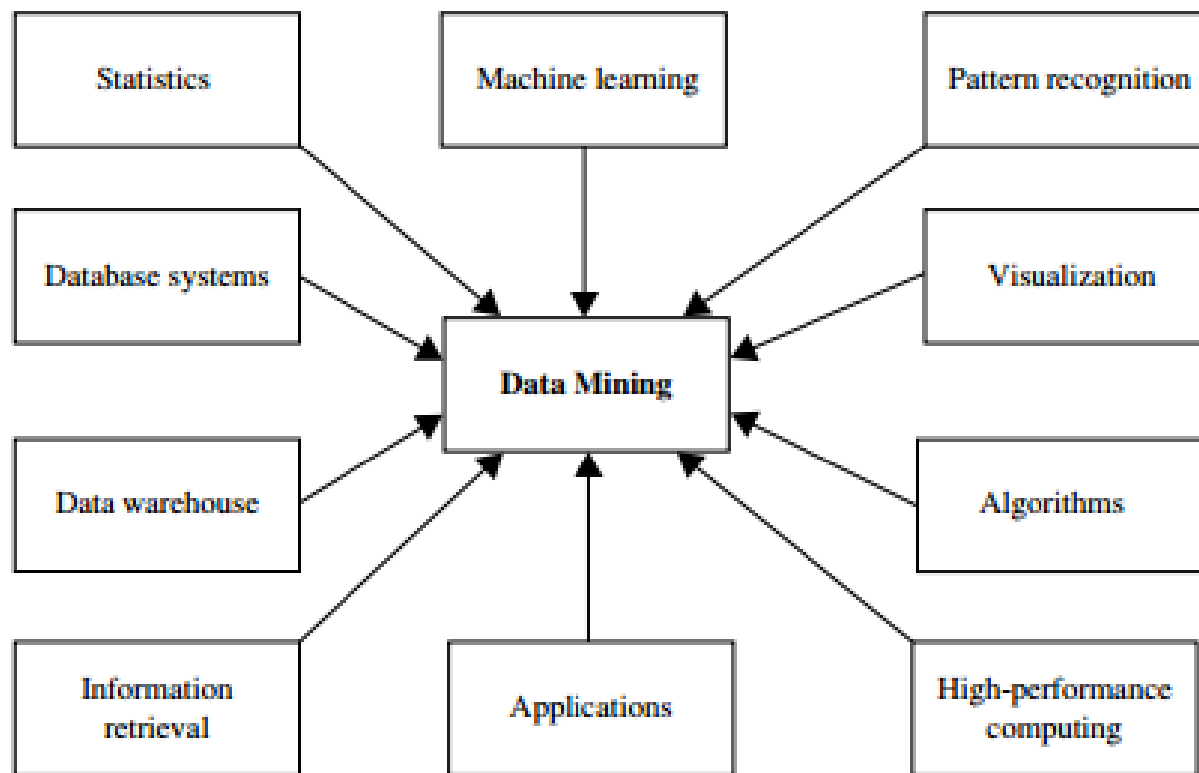


Figure 1.11 Data mining adopts techniques from many domains.



End of Lecture

Good Luck!

See you
in next lecture...

